

## Career Summary

- I have completed my **PhD** in the Department of **ECE at UCF** in **December 2024** with a concentration in **ML/AI**.
- I have **5+ years** of experience in **Designing, Developing, and implementing** Deep learning models like **LLMs, Transformers, GNNs, Generative models, Diffusion models, Foundation models, Multimodal Fusion models**, and **RL**, on various applications including **NLP, Bioinformatics, Healthcare, Forecasting, Electronics, CV**, and have familiarity with **version control(GIT), SQLite, MySQL, PostgreSQL, cloud computing(AWS), Parallel/Distributed computing (Multi GPU training), Docker, RAG, Prompt Engineering, MCP, and Data curation and Processing (Pandas, SQL, PowerBI, Kafka, Spark, Hadoop, Tableau)**.
- I have **6 years** of expertise in the **Python (NumPy, SciPy, Pandas, Regular Expression(Regex)), java, and C++** programming languages using **TensorFlow, Pytorch, CUDA, HuggingFace, Scikit-learn** frameworks, along with a deep understanding of **statistics and linear algebra**, and proficiency in clean, scalable, and documented coding practices.
- I have multiple publications in **top ML journals and conferences**.
- I also have industry experience during my internship as a **Data Scientist at Bayer** as well as working on joint projects with **Tokyo Electron, ARM, and Synopsys** on real-world ML applications and large scale applied ML projects.

## Experience

### Bayer Research and Development Services

Missouri, USA

Remote Sensing Data Science Intern

May 2023 - Aug 2023

Project:

- Determining Phenotype of corn plants from UAV (Unmanned Ariel Vehicle) images of corn fields.
- Implemented various CNN and transformer-based vision algorithms and evaluated and compared their performance to identify the best-suited model for the application.

### University of Central Florida (UCF)

Florida, USA

Research Assistant

Aug 2019 - Present

Primary Responsibility includes working on various ML related research Projects. Some of them are:

- KITE-DDI**: A Knowledge graph integrated Transformer architecture for accurately predicting Drug-Drug Interaction events from SMILES
- ProtEC**: A Transformer based deep learning model to accurately predict Enzyme Commission numbers from full-scale protein sequences
- ProtGO**: A Transformer based fusion model for accurately predicting Gene Ontology(GO) terms from full scale protein sequences.
- Collaborative Project with Tokyo Electron Ltd. to develop Design Flow for 3D IC design

## Selected Research Projects

### KITE-DDI

UCF, USA

A Knowledge graph integrated Transformer model for accurately predicting Drug-Drug Interaction Events

- A Knowledge Graph and transformer based fusion model have been proposed which could predict DDI events with SOTA accuracy.
- Shows robust performance with little dependency on sequence length and training set size.
- Paper Link: <https://ieeexplore.ieee.org/document/10909071>
- GitHub Repository: <https://github.com/azwad-tamir/KiteDDI>

### ProtEC

UCF, USA

A Transformer Based Deep Learning System for Accurate Annotation of Enzyme Commission Numbers

- A Bert-based Transformer model to predict all four Enzyme Commission numbers from full-scale Protein sequences.
- Pretrained ProtBert modules were selectively finetuned on the Uniprot SwissProt protein database to achieve SOTA accuracy.
- GitHub Repository: [https://github.com/azwad-tamir/Prot\\_EC](https://github.com/azwad-tamir/Prot_EC)
- Paper Link: <https://ieeexplore.ieee.org/document/10238761>

### ProtGO

UCF, USA

A Transformer Based Fusion model for accurately predicting Gene Ontology (GO) Terms from full-scale protein sequences.

- A Bert-based Transformer model to annotate full scale protein sequences with Gene Ontology (GO) terms using a classification type parallel ML fusion architecture.
- The proposed model is able to predict GO terms with better accuracy compared to other state-of-the-art methods and could understand motifs and protein 2D structures.
- GitHub Repository: <https://github.com/azwad-tamir/ProtGO>
- Paper Link: <https://arxiv.org/abs/2412.05776>

## Skin Cancer Detection

UCF, USA

### Medical Imaging using ML

- Several Deep Learning classification Model was implemented to detect skin cancer from images of skin lesions.
- Transfer Learning was used on different Vision Transformer (ViT) models to classify skin cancer types on the HAM10000 dataset.
- ViT-based models showed significant improvement over the CNN models.
- GitHub Repository: [https://github.com/azwad-tamir/Transfer\\_ViT](https://github.com/azwad-tamir/Transfer_ViT)

## Education

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### University of Central Florida (UCF)

Orlando, Florida, USA

#### PhD in Electrical Engineering

Fall 2019 - Fall 2024

- **Research Areas:** Developing various Deep-learning Models in NLP, CV, and GNN for low data Applications in various real-world applications.
- **Dissertation Title:** Architectural and Training regime Modifications to Transformer models for Low Data Applications
- **Current CGPA:** 4.0 out of 4.0
- **Courses:** Machine Learning, Machine Learning Methods for Biomedical Data, Computational Genomics, Data Mining Methodology II, Intro to Neural Networks, Computer Vision, 3D Computer Vision, Design & Analysis of Algorithms, CMOS Analog & Digital Circuit Design, Malware Software Vulnerability, Incident Response Technologies

### University of Central Florida (UCF)

Florida, USA

#### Masters of Science in Electrical Engineering

Aug 2019 - Spring 2023

- **CGPA:** 4.00 out of 4.00
- **Major:** Electrical Engineering

## Selected Publications

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A. Tamir and J. -S. Yuan, "KITE-DDI: A Knowledge Graph Integrated Transformer Model for Accurately Predicting Drug-Drug Interaction Events From Drug SMILES and Biomedical Knowledge Graph," in IEEE Access, vol. 13, pp. 40028-40043, 2025, doi: 10.1109/ACCESS.2025.3547594.

A. Tamir, M. Salem and J. -S. Yuan, "ProtEC: A Transformer Based Deep Learning System for Accurate Annotation of Enzyme Commission Numbers," in IEEE/ACM Transactions on Computational Biology and Bioinformatics, vol. 20, no. 6, pp. 3691-3702, Nov.-Dec. 2023.

A. Tamir and J.-S. Yuan, "ProtGO: A Transformer based Fusion Model for accurately predicting Gene Ontology (GO) Terms from full scale Protein Sequences." arXiv, 2024. doi: 10.48550/ARXIV.2412.05776. Available: <https://arxiv.org/abs/2412.05776>

A. Tamir, E. Watson, B. Willett, Q. Hasan and J. Yuan, "Crime Prediction and Forecasting using Machine Learning Algorithms," International Journal of Computer Science and information Technologies, Vol. 12(2), 2021, pp. 26-33, ISSN:0975-9646.

A. Tamir, C. Jahan, M. Saif, S. Zaman, M. Islam, A. Khan, S. Fattah, and C. Shahnaz, "Detection of anemia from image of the anterior conjunctiva of the eye by image processing and thresholding," 2017 IEEE Region 10 Humanitarian Technology Conference (R10-HTC), Dhaka, 2017, pp. 697-701. doi: 10.1109/R10-HTC.2017.8289053.

## Skills

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|--------------------|----------------------------------------------------------------------------------------------------------------------------|
| <b>Programming</b> | Python (NumPy, ProtoBuf, Pandas, etc.), R, C/C++, Bash, Java, Matlab/Simulink, Verilog, SQL, PowerBI, Kafka, Spark, Hadoop |
| <b>Frameworks</b>  | HuggingFace, PyTorch, TensorFlow, Scikit-learn, Keras, AWS                                                                 |
| <b>Software</b>    | Agile, Synopsys Design & ICC2 Compiler, Lumerical FDTD solutions, Cadence Virtuoso, SPICE, Proteus                         |
| <b>Hardware</b>    | Embedded Systems, Robotics, Microcontroller (AVR), Arduino, Raspberry Pi, FPGA                                             |

## Certifications

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|----------------------------|----------------------------------------------------------------------------------------|
| <b>Stanford University</b> | Machine Learning, course conducted by Coursera                                         |
| <b>DeepLearning.AI</b>     | Improving Deep Neural Networks: Hyperparameter Tuning, Regularization and Optimization |
| <b>DeepLearning.AI</b>     | Neural Networks and Deep Learning                                                      |
| <b>DeepLearning.AI</b>     | Natural Language Processing with Attention Models                                      |
| <b>DeepLearning.AI</b>     | Structuring Machine Learning Projects                                                  |
| <b>Linkedin Learning</b>   | SQL Essentials Training                                                                |
| <b>UIUC</b>                | VLSI CAD Part II: Layout. Course conducted by Coursera                                 |